

YASAS MANUJAYA EKANAYAKE

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Negombo, Sri Lanka

RESEARCH INTERESTS

- Artificial Intelligence and Machine Learning
- Database Security and Query Optimization
- Intelligent Systems, Intelligent Computing and Multi-Agent Systems
- Software System Security and Cybersecurity

EDUCATION

- **University of Moratuwa** 2021 – 2025
B.Sc. (Hons) in Information Technology Sri Lanka
 - **CGPA: 3.82/4.20** | Dean's List (4x) | [Transcript](#)
 - Courses: Artificial Intelligence, Machine Learning, Data Security, Distributed Systems, Software Engineering, Computer Science, Computational Mathematics, Computational Statistics
- **Richmond College Galle** 2019
GCE Advanced Level Sri Lanka
 - **Combined Maths: A, Chemistry A, Physics B** | Z-Score: 1.7632 | [Results](#)

LANGUAGES TEST SCORES

IELTS (18 June, 2026) | Overall Band Score: 7.5 | CEFR Level C1 | [Test Score](#)

EXPERIENCE

- **University of Moratuwa**  Jul 2026 – Present
Visiting Lecturer - BIT(External) Moratuwa, Sri Lanka
 - Design and deliver comprehensive lectures and coursework of Cloud Computing module.
- **Informatics Institute of Technology**  Jun 2026 – Present
Final Year Project Supervisor Colombo, Sri Lanka
 - Direct final-year undergraduate research projects within the School of Computing.
- **University of Moratuwa**  Nov 2025 – Present
Lecturer Moratuwa, Sri Lanka
 - Working as a Lecturer in Computer Science and Engineering Department.
 - Working in collaborative researches.
 - Mentoring, Evaluating, Supervising undergraduate students in their projects.
- **WSO2 LLC**  Jan 2024 – Jul 2024
Software Engineer Intern Colombo, Sri Lanka
 - Worked with MI and R&D teams on AI-integrated mediators and MI VSCode extension.
 - Implemented proxy, message store, and local entry artifacts with dynamic form validation.
 - Conducted testing, technical documentation, and presentations to stakeholders.

RESEARCH PROJECTS

- **Evaluating Script & Tokenization Impact on Low-Resource LLMs - Supervised** 2026
Tools: Python, Pytorch, LLM, NLP
 - Conducted an empirical study on 31 open-source Large Language Models (350M–14.7B) evaluating Sinhala script vs. Romanized Sinhala (Singlish) performance across 9,470+ evaluation items.
 - Demonstrated that a reported 312-fold performance gap between scripts was largely a tokenization artifact, decomposing the loss into tokenizer normalizer vs. real model degradation using bits-per-byte (BPB).
 - Re-evaluated 11 instruction-tuned models across tasks (MMLU, offensive language detection, PIQA) and built a deterministic Python grapheme-level transliterator to evaluate zero-shot cross-script performance.
- **Quantifying The Trade-Off Between De-Identification Perturbation And Re-Identification Risk In ECG Signals** 2026 - Present
Tools: Python, Pytorch, LLM, NLP
 - This research proposes a comprehensive analytical study to quantify the relationship between perturbation severity introduced by different ECG de-identification techniques and the accuracy of re-identification attacks, with the goal of identifying practical thresholds beyond which re-identification becomes infeasible while preserving medical application.
- **Hybrid Sentiment Analysis with Explainable AI (XAI) for mental health detection on social media** 2025 - 2026
Tools: Python, Pytorch, NLP, DistilBERT, LIME
 - Contributed to a research project built on transformer-based architectures, integrating DistilBERT with cross-attention mechanisms and curating custom datasets with MBTI labels.
 - Implemented adversarial training and designed cross-attention architectures, leveraging DistilBERT with LIME and rule based system.
- **Detection System for Online Safety Against Cyber Threats and Social Engineering Attacks (FYP)** 2024 - 2025
Tools: Python, Pytorch TensorFlow, Keras, NLP
 - An intelligent detection system to identify and mitigate manipulative behavior and fraudulent activity in real-time conversations.
 - The project targeted social engineering attacks conducted through chat and audio messages by both humans and web bots, leveraging cutting-edge AI to safeguard users.
 - Contributed to design and implement a multi-layered detection framework combining Natural Language Processing (DistilBERT with adversarial training), rule-based systems

SOFTWARE PROJECTS

- **Bus-Ease: Intelligent Route & Seat Reservation System** 2025
Tools: React.js, Node.js, MongoDB, Stripe, Vercel [🔗]
 - Built platform for highway bus search and booking with real-time seat updates and secure payments.
 - Contributed intelligent route planning algorithm and full-stack development.
- **AI-Powered Resume Search Chatbot** 2025
Nvidia NIM, LangChain, ChromaDB vector database, Streamlit, GitHub, Python [🔗]
 - AI-Powered Resume Search Chatbot using Retrieval Augmented Generation (RAG) and NVIDIA NIM microservices to revolutionize candidate screening.
 - The system uses NVIDIA NeMo Retriever for semantic embeddings and Llama 3.1 70B for natural language generation, enabling hiring managers to search resumes conversationally instead of using rigid keyword filters
- **Intelligent Document Process Mediator for WSO2 MI** 2024
Java, Maven, WSO2 MI, ChatGPT Vision API
 - Implemented a mediator to extract structured JSON from PDFs and images using AI model integration. Reduced the manual processing by 60%.

PUBLICATIONS

[1] Ekanayake, Yasas Manujaya, and Lochandaka Ranathunga. "Hybrid Sensing Approach for Real-Time Chat Behavior Detection and Classify Social Engineered Attacks." 2025 IEEE 19th International Conference on Industrial and Information Systems (ICIIS), 16 Jan. 2026, pp. 19–24, <https://doi.org/10.1109/iciis69028.2026.11450818>. Accessed 9 Apr. 2026.

SKILLS

- **Programming Languages:** Python, Java, C, C++, JavaScript, SQL
- **Web Technologies:** NodeJS, ReactJS, Redux, Redis, Cloud
- **Database Systems:** MongoDB, MSSQL, ChromaDB
- **Data Science & Machine Learning:** Torch, Keras, NLP, ML Algorithms, Pandas, Spark, FAISS, XAI
- **Cloud Technologies:** Oracle, AWS
- **Frameworks, Tools & Version Control:** Git, Github, Spring Boot, Docker, Jenkins, Flutter
- **Other Tools & Technologies:** Selenium, Cucumber, Serenity BDD, Rest Assured, Vercel, Streamlit, Figma

CERTIFICATIONS

- **Building Transformer-Based Natural Language Processing Application | NVIDIA:** [Certificate](#) 2026
- **Fundamentals of Accelerated Data Science | NVIDIA:** [Certificate](#) 2026
- **Building RAG Agents with LLMs | NVIDIA:** [Certificate](#) 2026
- **Supervised Machine Learning: Regression and Classification | DeepLearning.AI:** [Certificate](#) 2026

ACADEMIC CONTRIBUTION

Reviewer

- NeurIPS 2026 GlobalSouthAI Reviewer , NeurIPS
- MerCon 2026 : Data Science and AI Track; Software Engineering Track , University of Moratuwa
- ADsCAI 2026 : Applied AI & Data Science for Industry , University of Moratuwa
- FYP Poster Judge 2026 University of Moratuwa

Poster & Presentation Judge

- ADsCAI 2026 Poster Presentation , University of Moratuwa
- Final Year Project Poster Presentation Judge 2026 , University of Moratuwa

HONORS AND ACTIVITIES

- **Dean’s List (4 times)** 2021 – 2025
University of Moratuwa
- **Batch Representative – Faculty of IT** 2023
University of Moratuwa
- **Vice President – Southern Youth Council** 2022
Sri Lanka Youth Association

COMPETITIONS

- **RealHack 4.0 - Hackathon** 2023
Top 50 among 1000+ Teams organized by Hackerrank

ADDITIONAL INFORMATION

Languages: English (Fluent), Sinhala (Native), Tamil(Basic)

Interests: Cricket, Robotics, Games

REFERENCES

1. **Prof. Sanath Jayasena** | [Profile](#)

Associate Professor , Department of Computer Science Engineering, Faculty of Engineering, University of Moratuwa, Sri Lanka

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2. **Dr. Lochandaka Ranathunga** | [Profile](#)

Senior Lecturer , Department of Information Technology, Faculty of Information Technology, University of Moratuwa, Sri Lanka

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3. **Dr. A.L.A.R.R. Thanuja (B.Sc. (Sri Lanka), Ph.D.(USA))** | [Profile](#)

Senior Lecturer, Department of Computational Mathematics, Faculty of Information Technology. University of Moratuwa, Sri Lanka.

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